

Name: Sagi Sahasra Srihasa

Roll no: 2520030008

Step 1:

```
1  /* Name: Sagi Sahasra Srihasa
2     Roll no: 2520030008
3  */
4
5  -- Step 1: Create Physician Table
6  ● CREATE DATABASE physician_db;
7  ● USE physician_db;
8  ● CREATE TABLE physician (
9      employeeid INT PRIMARY KEY,
10     name VARCHAR(100),
11     position VARCHAR(100),
12     ssn VARCHAR(20)
13 );
```

✓	1 17:51:31	CREATE DATABASE physician_db	1 row(s) affected
✓	2 17:51:34	USE physician_db	0 row(s) affected
✓	3 17:51:36	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100...	0 row(s) affected

Step 2:

```
-- Step 2: Insert Data into Physician
INSERT INTO physician VALUES
(1,'Sahasra','Staff Internist','111111111'),
(2,'Ravinder','Attending Physician','222222222'),
(3,'Srilatha','Surgical Attending Physician','333333333'),
(4,'Santhvin','Senior Attending Physician','444444444'),
(5,'Varshika','Head Chief of Medicine','555555555'),
(6,'Manu','Surgical Attending Physician','666666666'),
(7,'Shiny','Surgical Attending Physician','777777777'),
(8,'Pranitha','MD Resident','888888888'),
(9,'Angela','Attending Psychiatrist','999999999');

-- Step 3: Create Department Table
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);

-- Step 4: Insert Data into Department
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
```

✓	4	17:52:17	INSERT INTO physician VALUES (1,'Harshika','Staff Internist','111111111'), (2,'Ravinder','Attending Physician','222...	5
✓	5	17:52:22	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	6
✓	6	17:52:26	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	7

Step 3:

```

38      -- Step 5: INNER JOIN
39      • SELECT d.name AS Department,
40             p.name AS Head_Physician
41      FROM department d
42      JOIN physician p
43      ON d.head = p.employeeid;
44      -- Step 6: WHERE Clause

```

Result Grid   Filter Rows: Export:  Wrap (

	Department	Head_Physician
▶	General Medicine	Santhvin
	Surgery	Shiny
	Psychiatry	Angela

Step 4:

```
44      -- Step 6: WHERE Clause
45 •    SELECT *
46      FROM physician
47      WHERE position = 'Surgical Attending Physician';
48
49
```

	employeeid	name	position	ssn
▶	3	Srilatha	Surgical Attending Physician	333333333
	6	Manu	Surgical Attending Physician	666666666
	7	Shiny	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

Step 5:

```
51      -- Step 7: WHERE + LIKE
52 •    SELECT *
53      FROM physician
54      WHERE name LIKE 'John%';
55
56
```

	employeeid	name	position	ssn
*	NULL	NULL	NULL	NULL

Step 6:

```
59      -- Step 8: COUNT()
60 •    SELECT COUNT(*) AS Total_Physicians
61      FROM physician;
62
```

	Total_Physicians
▶	9

Step 7:

```
65      -- Step 9: COUNT DISTINCT
66 •    SELECT COUNT(DISTINCT position) AS Different_Positions
67      FROM physician;
68
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Different_Positions			
▶ 7			

Step 8:

```
70      -- Step 10: GROUP BY
71 •    SELECT position,
72          COUNT(*) AS Total_Employees
73      FROM physician
74      GROUP BY position;
75
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
position	Total_Employees		
▶ Staff Internist	1		
Attending Physician	1		
Surgical Attending Physician	3		
Senior Attending Physician	1		
Head Chief of Medicine	1		
MD Resident	1		
Attending Psychiatrist	1		

Step 9:

```
77 -- Step 11: GROUP BY + HAVING
78 • SELECT position,
79       COUNT(*) AS Total_Employees
80 FROM physician
81 GROUP BY position
82 HAVING COUNT(*) > 1;
83
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
position	Total_Employees		
Surgical Attending Physician	3		

Step 10:

```
85 -- Step 12: ORDER BY ASC
86 • SELECT *
87 FROM physician
88 ORDER BY name ASC;
89
90
```

Result Grid	Filter Rows:	Edit:	Export/Imp
employeeid	name	position	ssn
9	Angela	Attending Psychiatrist	999999999
1	Harshika	Staff Internist	111111111
6	Manu	Surgical Attending Physician	666666666
8	Pranitha	MD Resident	888888888
2	Ravinder	Attending Physician	222222222
4	Santhvin	Senior Attending Physician	444444444
7	Shiny	Surgical Attending Physician	777777777
3	Srilatha	Surgical Attending Physician	333333333
5	Varshika	Head Chief of Medicine	555555555
NULL	NULL	NULL	NULL

Step 11:

```
93      -- Step 13: ORDER BY DESC
94      SELECT *
95      FROM physician
96      ORDER BY employeeid DESC;
97
98
```

Result Grid	Filter Rows:	Edit:	Export/Im
employeeid	name	position	ssn
9	Angela	Attending Psychiatrist	999999999
8	Pranitha	MD Resident	888888888
7	Shiny	Surgical Attending Physician	777777777
6	Manu	Surgical Attending Physician	666666666
5	Varshika	Head Chief of Medicine	555555555
4	Santhvin	Senior Attending Physician	444444444
3	Srilatha	Surgical Attending Physician	333333333
2	Ravinder	Attending Physician	222222222
1	Harshika	Staff Internist	111111111
NULL	NULL	NULL	NULL

Step 12:

```
101      -- Step 14: WHERE + IN
102      SELECT *
103      FROM physician
104      WHERE employeeid IN
105      (
106          SELECT head
107          FROM department
108      );
109
```

Result Grid	Filter Rows:	Edit:	Export/Import:
employeeid	name	position	ssn
4	Santhvin	Senior Attending Physician	444444444
7	Shiny	Surgical Attending Physician	777777777
9	Angela	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Step 13:

```
112 -- Step 15: NOT IN
113 • SELECT *
114 FROM physician
115 WHERE employeeid NOT IN
116 (
117     SELECT head
118     FROM department
119 );
120
```

Result Grid				
Filter Rows:				
Edit:				
Export/Imp				
	employeeid	name	position	ssn
▶	1	Harshika	Staff Internist	111111111
	2	Ravinder	Attending Physician	222222222
	3	Srilatha	Surgical Attending Physician	333333333
	5	Varshika	Head Chief of Medicine	555555555
	6	Manu	Surgical Attending Physician	666666666
	8	Pranitha	MD Resident	888888888
★	NULL	NULL	NULL	NULL

Step 14:

```
122
123 -- Step 16: GROUP BY + ORDER BY
124 • SELECT position,
125         COUNT(*) AS Total_Employees
126 FROM physician
127 GROUP BY position
128 ORDER BY Total_Employees DESC;
129
```

Result Grid		
Filter Rows:		
Export:		
Wrap Cell Content:		
	position	Total_Employees
▶	Surgical Attending Physician	3
	Staff Internist	1
	Attending Physician	1
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Step 15;

```
131      -- Step 17: WHERE + LIKE
132 •    SELECT *
133      FROM physician
134      WHERE position LIKE '%Attending%';
135      -- Step 18: JOIN + WHERE
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	employeeid	name	position	ssn
▶	2	Ravinder	Attending Physician	222222222
	3	Srilatha	Surgical Attending Physician	333333333
	4	Santhvin	Senior Attending Physician	444444444
	6	Manu	Surgical Attending Physician	666666666
	7	Shiny	Surgical Attending Physician	777777777
	9	Angela	Attending Psychiatrist	999999999
✱	NULL	NULL	NULL	NULL

Step 16;

```
135      -- Step 18: JOIN + WHERE
136 •    SELECT p.name
137      FROM physician p
138      JOIN department d
139      ON p.employeeid = d.head
140      WHERE d.name = 'Surgery';
141
```

Result Grid		Filter Rows:	Export:	Wrap
	name			
▶	Shiny			

Step 17:

```
144      -- Step 19: JOIN + ORDER BY
145 •    SELECT d.name AS Department,
146           p.name AS Head_Physician
147      FROM department d
148     JOIN physician p
149     ON d.head = p.employeeid
150     ORDER BY d.name;
151      -- Step 20: WHERE + NOT LIKE
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Department	Head_Physician			
▶	General Medicine	Santhvin			
	Psychiatry	Angela			
	Surgery	Shiny			

Step 18:

```
151      -- Step 20: WHERE + NOT LIKE
152 •    SELECT *
153     FROM physician
154     WHERE position NOT LIKE '%Surgical%';
155
```

Result Grid					Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	employeeid	name	position	ssn				
▶	1	Harshika	Staff Internist	111111111				
	2	Ravinder	Attending Physician	222222222				
	4	Santhvin	Senior Attending Physician	444444444				
	5	Varshika	Head Chief of Medicine	555555555				
	8	Pranitha	MD Resident	888888888				
	9	Angela	Attending Psychiatrist	999999999				
*	NULL	NULL	NULL	NULL				

Step 19:

```
158 -- Step 21: COUNT + HAVING
159 • SELECT position,
160       COUNT(*) AS Total
161 FROM physician
162 GROUP BY position
163 HAVING COUNT(*) >= 2;
164
165 -- Step 22: EXISTS
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	position	Total
▶	Surgical Attending Physician	3

Step 20:

```
165 -- Step 22: EXISTS
166 • SELECT *
167 FROM department d
168 WHERE EXISTS
169 (
170   SELECT 1
171   FROM physician p
172   WHERE p.employeeid = d.head
173 );
174 -- Step 23: MAX()
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
•	NULL	NULL	NULL

Step 21:

```
174 -- Step 23: MAX()
175 • SELECT MAX(employeeid) AS Highest_Employee_ID
176 FROM physician;
177 -- Step 24: MIN()
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Highest_Employee_ID
▶	9

Step 22:

```
177 -- Step 24: MIN()
178 • SELECT MIN(employeeid) AS Lowest_Employee_ID
179 FROM physician;
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	Lowest_Employee_ID
▶	1

Step 15: